SIT720 8.1D

Sydney Housing Price Prediction and Decision Support System

Machine Learning Mini Project - Distinction Task

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Unit: SIT720 - Machine Learning

Dataset: 105 sold Sydney properties across three suburbs

Deployment: Gradio web application

Executive summary

This project develops a reproducible machine-learning decision-support system for Sydney residential property prices. A manually assembled dataset of 105 sold properties was constructed from public Realestate.com.au sold-listing pages, with 35 observations each from Bondi Beach, Parramatta and Blacktown. These suburbs deliberately span a premium coastal market, a major western-centre market and a lower-priced suburban market. The workflow covers data quality assessment, exploratory analysis, feature engineering, three regression approaches, training-only cross-validation, held-out testing, error analysis, comparison with LLM and manual estimates, and a deployable Gradio prototype.

Random Forest was selected using the lowest five-fold cross-validation MAE on the training set (A$471,511). On the untouched test set it achieved MAE A$509,178, RMSE A$914,559 and R2 0.803. The largest errors occurred in Bondi Beach, showing that structured listing fields do not fully capture prestige, renovation, views, micro-location and atypical property characteristics. The deployment is therefore presented as decision support rather than a professional valuation.

1. Problem definition and data collection

The prediction target is the final sold price in Australian dollars. Each row contains suburb, address, sold price, bedrooms, bathrooms, parking, land size, property type and sale date. The source-page URL is also stored for provenance. The three suburbs were chosen to create substantial market contrast. In this collected sample, the median sold price is A$4.105 million in Bondi Beach, A$1.250 million in Parramatta and A$1.080 million in Blacktown. This spread makes the modelling task more informative than sampling a single narrow price segment.

Suburb	n	Median	Mean	Minimum	Maximum
Blacktown	35	A$1,080,000	A$1,149,086	A$820,000	A$1,900,000
Bondi Beach	35	A$4,105,000	A$4,170,230	A$840,000	A$9,400,000
Parramatta	35	A$1,250,000	A$1,329,368	A$720,000	A$2,300,000

Data quality is the main constraint. The sample is manually collected and is not a probability sample of all Sydney sales. Parking is missing in 13 rows and land size in 38 rows, while the target has no missing values. Numeric missing values are handled inside the modelling pipeline using median imputation, which is more robust than the mean for skewed property attributes. Categorical variables are imputed with the most frequent category and one-hot encoded. The records also omit consistently measured floor area, property age, renovation quality, views, street frontage, school catchment, zoning and exact geospatial accessibility. These omissions are important sources of residual error and potential bias.

2. Data understanding and feature engineering

Before creating engineered features, I expected suburb, property type and bedrooms to be the three strongest predictors. Suburb represents location and market segment; property type separates detached from attached dwellings; and bedroom count is a simple proxy for usable size. Land size was also expected to matter, but it is unavailable for many listings. The observed data support the importance of location: the suburb boxplot shows a much higher and wider Bondi Beach price range than the western suburbs.

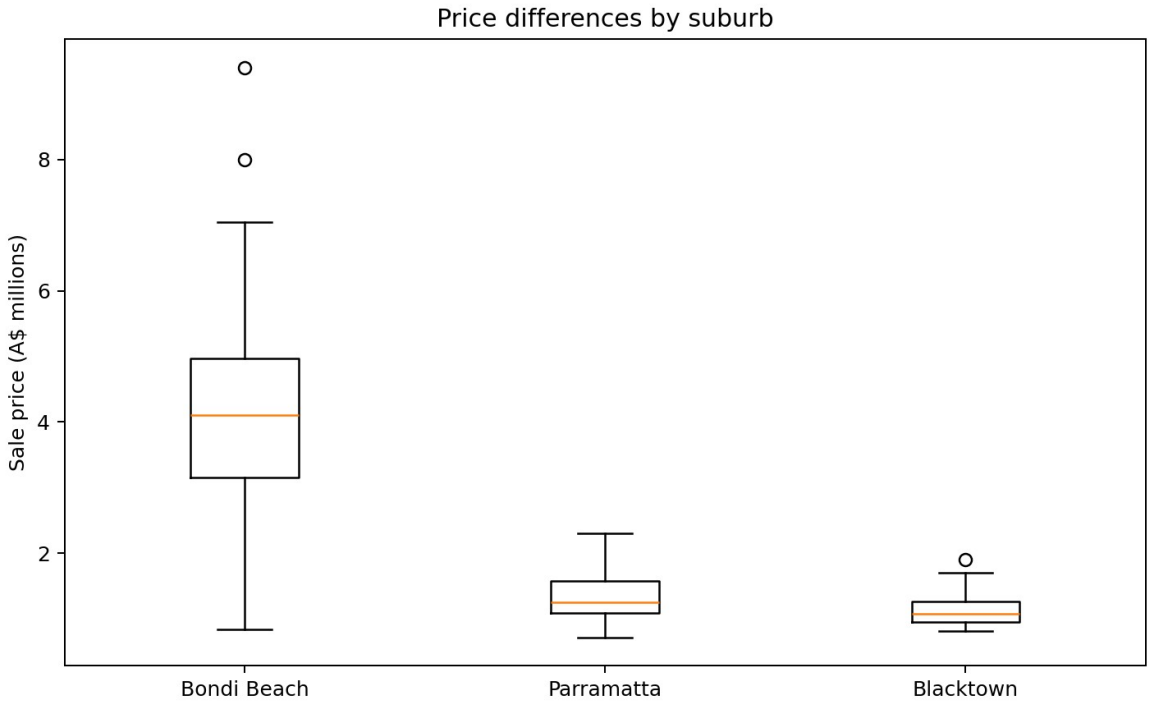


Figure 1. Sold-price differences across the three selected suburbs.

The overall price distribution is strongly right-skewed because a small number of premium Bondi Beach properties extend above A$5 million. Land size has a positive relationship with price within the western suburbs, but the combined plot also shows why land size cannot be interpreted without location: similarly sized blocks in Bondi Beach sell at very different price levels.

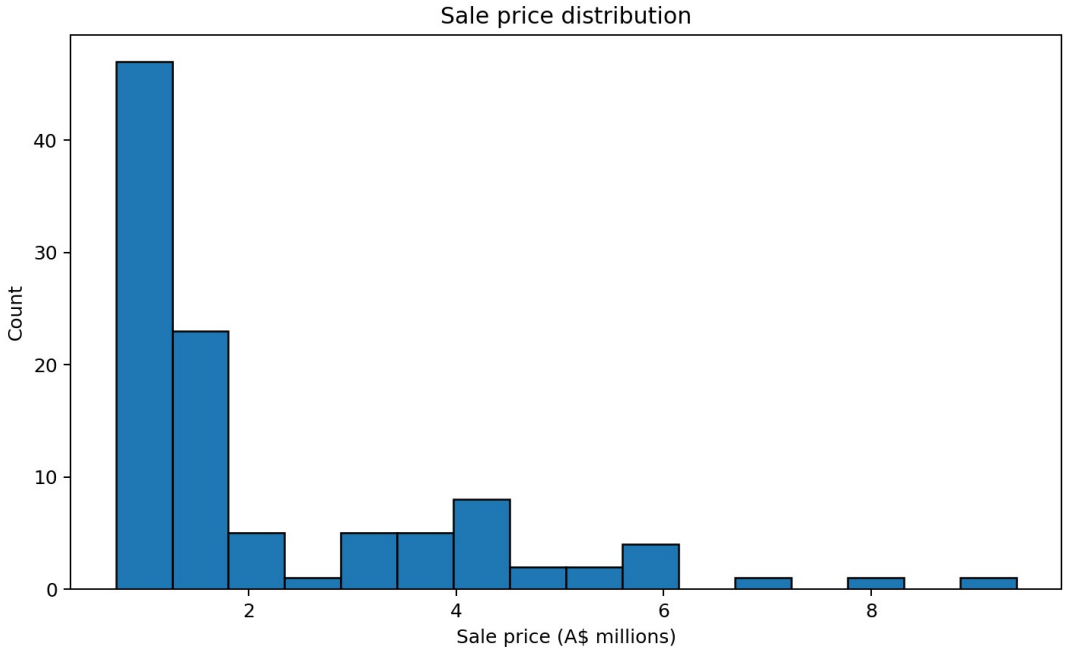


Figure 2. Distribution of sale prices in the 105-property sample.

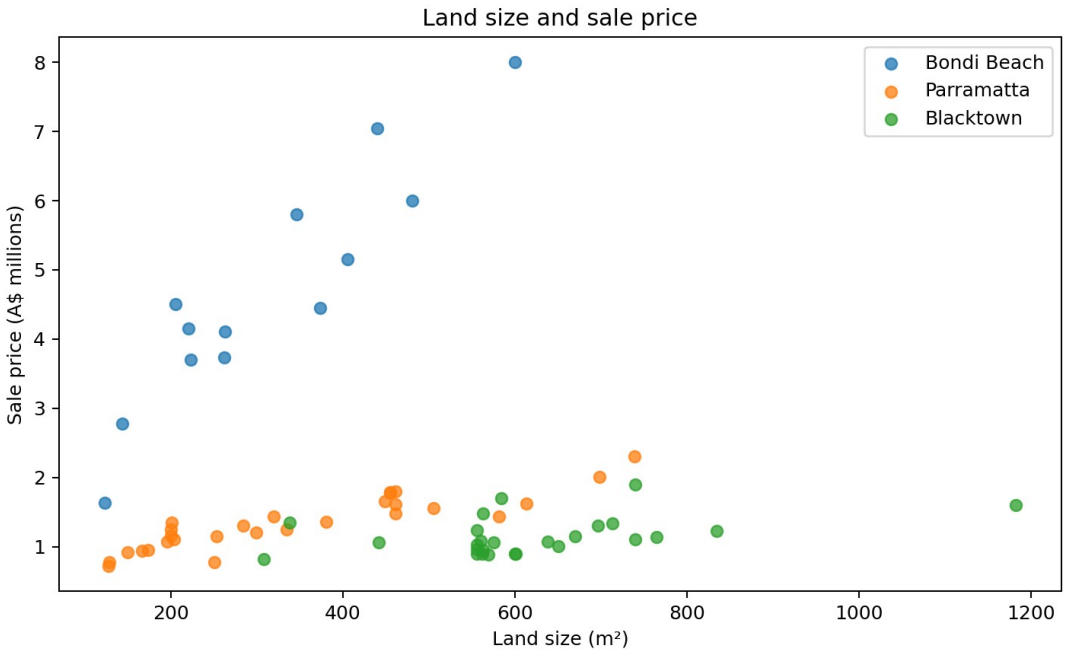


Figure 3. Land size versus sold price, separated by suburb.

Feature engineering added sale year, month and quarter, total bedrooms plus bathrooms, land size per bedroom and parking spaces per bedroom. These variables allow simple time effects and interactions between dwelling capacity and site size without discarding the original measurements. The observed yearly medians are descriptive only: coverage is sparse in earlier years for Parramatta and Blacktown, so the trend plot should not be treated as a population-level house-price index.

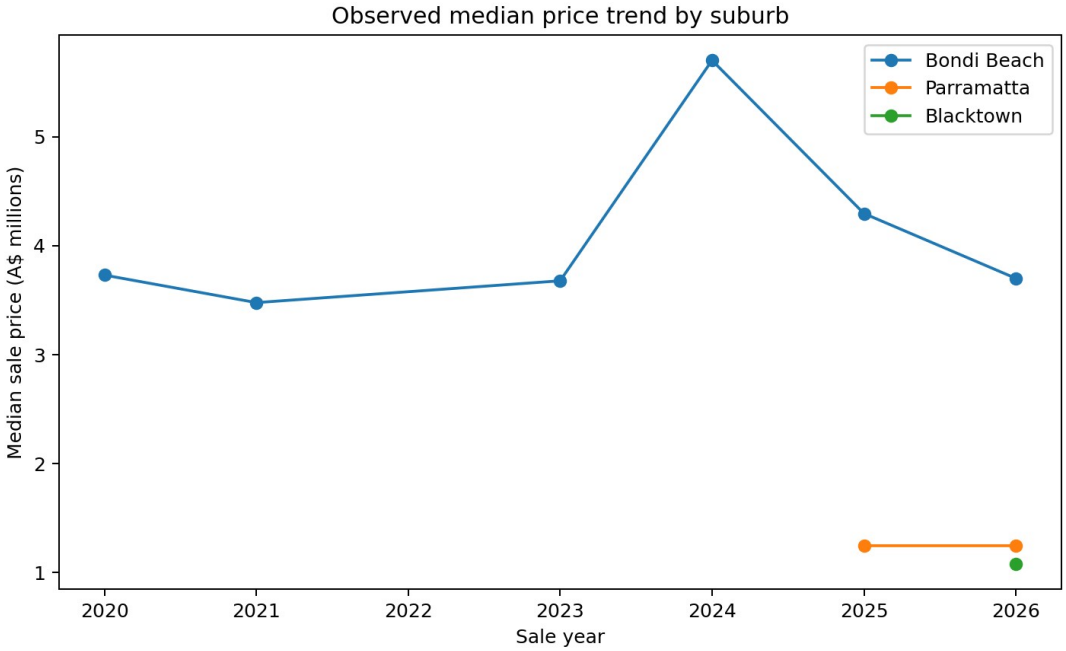


Figure 4. Observed median sale-price trend by suburb; sparse years should be interpreted cautiously.

3. Model development and evaluation

A single 80:20 hold-out split was created with suburb stratification. The 20% test set (21 properties, seven from each suburb) remained untouched during model and hyperparameter selection. Five-fold cross-validation on the training set was also stratified by suburb. The primary selection metric was mean absolute error (MAE), because it is directly interpretable in dollars and is less dominated by a few extreme sales than squared error.

Three approaches were compared. Ridge regression provides a regularised linear baseline and was expected to be stable but potentially underfit nonlinear relationships. Random Forest can capture nonlinear effects and interactions while reducing the variance of individual trees; before training, it was expected to perform best. Gradient Boosting sequentially corrects errors and can model complex relationships with relatively shallow trees, although aggressive settings can overfit a small dataset.

Model	CV MAE	CV RMSE	CV R2	Train MAE	Test MAE	Test R2
Random Forest	A$471,511	A$761,395	0.779	A$214,191	A$509,178	0.803
Gradient Boosting	A$487,554	A$771,893	0.771	A$260,003	A$449,500	0.821
Ridge	A$496,132	A$752,838	0.786	A$422,401	A$576,347	0.834

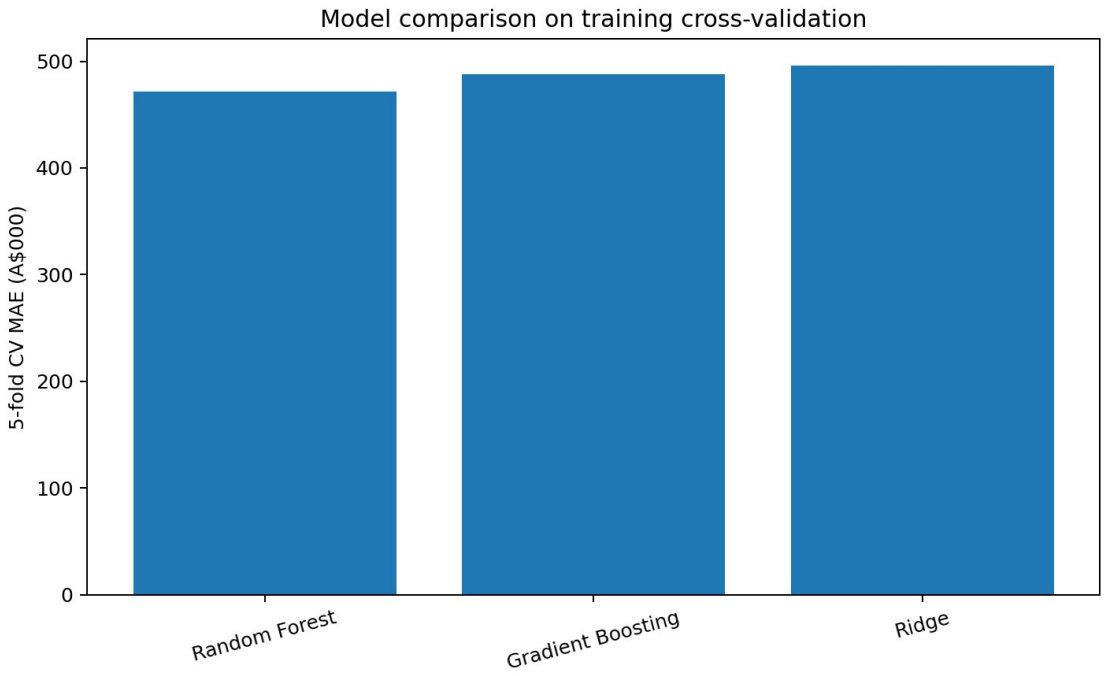


Figure 5. Five-fold training cross-validation MAE used for model selection.

Random Forest had the lowest cross-validation MAE (A$471,511) and was therefore selected before examining test performance. Its best grid-search setting used 200 trees, maximum depth 6 and minimum leaf size 1. Gradient Boosting produced the lowest held-out test MAE (A$449,500), while Ridge produced the highest held-out R2 (0.834). However, changing the selected model after seeing the test results would leak test information into model selection. The final recommendation therefore remains Random Forest because the decision rule was fixed on training-only cross-validation.

The complexity experiment provides evidence about bias and variance. Increasing Random Forest depth from 2 to 6 reduces cross-validation MAE from about A$526k to A$475k, but allowing unlimited depth continues to reduce training error without a comparable validation improvement. This is mild

overfitting pressure. Ridge has a much smaller training-validation gap but a higher MAE, consistent with a more constrained model that may underfit the nonlinear market structure.

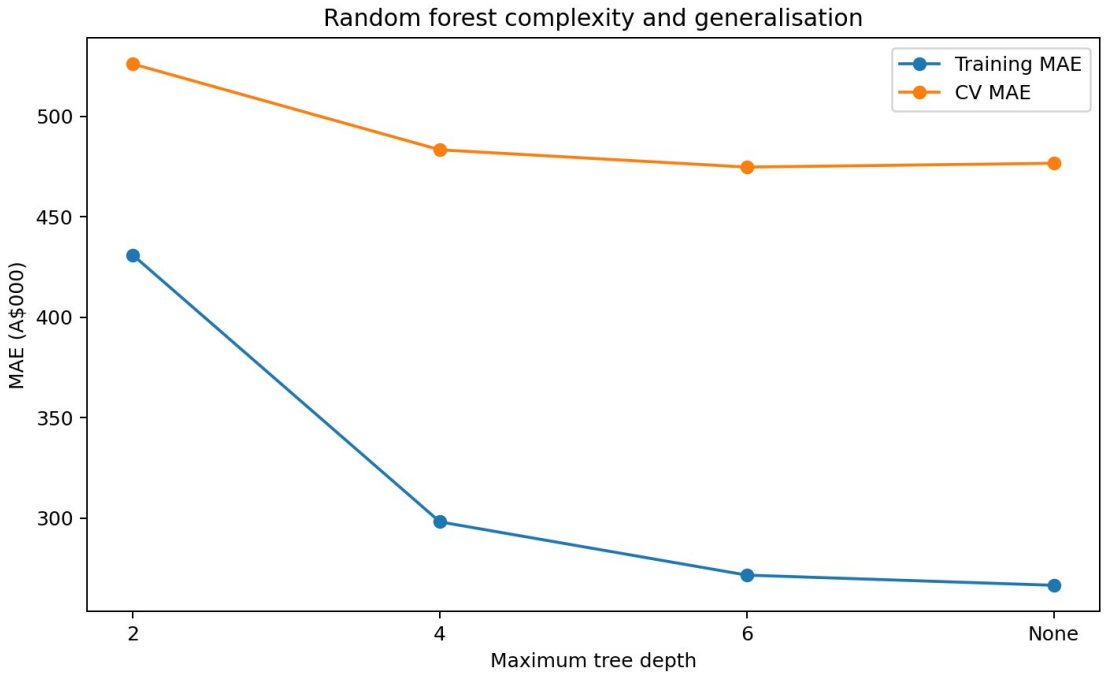


Figure 6. Random Forest depth, training error and validation error.

Random Forest feature	Importance
suburb_Bondi Beach	0.606
bedrooms	0.091
sale_month	0.074
parking	0.050
bed_bath_total	0.045
parking_per_bed	0.021
property_type_Other	0.020
land_per_bed	0.017
land_size_m2	0.017
sale_quarter_1	0.014

The Bondi Beach indicator dominates feature importance (0.606), followed by bedrooms (0.091), sale month (0.074), parking (0.050) and combined bed/bath capacity (0.045). This broadly supports the original expectation that location and size-related features dominate. Importances are model-specific and do not prove causal effects; correlated features can also split importance between them.

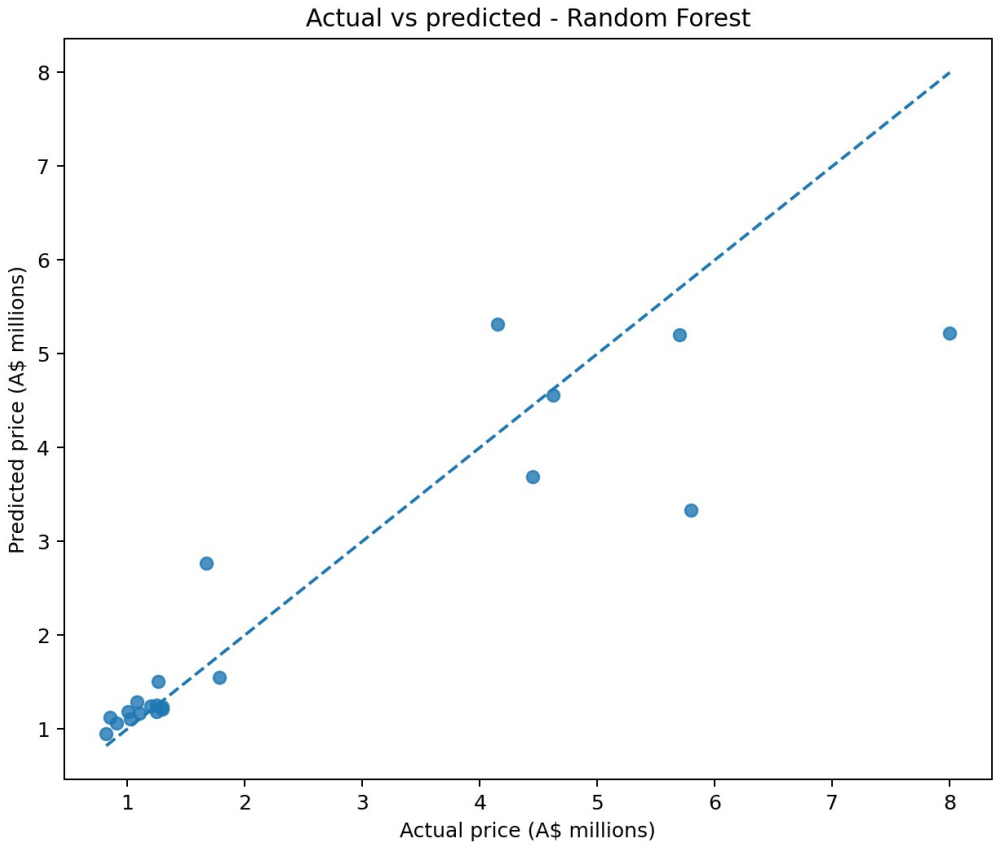


Figure 7. Held-out actual versus predicted prices for the selected Random Forest.

4. Investigating prediction failures

Property	Actual	Predicted	Abs. error	% error
10 Forest Knoll Avenue	A$8,000,000	A$5,221,356	A$2,778,644	34.7%
29 Sir Thomas Mitchell Road	A$5,800,000	A$3,330,239	A$2,469,761	42.6%
9 Cox Avenue	A$4,153,000	A$5,319,484	A$1,166,484	28.1%
4/22-28 O'Brien Street	A$1,670,000	A$2,770,971	A$1,100,971	65.9%
160 Hall Street	A$4,450,000	A$3,688,567	A$761,433	17.1%

All five largest test errors are in Bondi Beach. The two largest are under-predictions for 10 Forest Knoll Avenue (A$2.779 million error) and 29 Sir Thomas Mitchell Road (A$2.470 million error). The largest relative error is the attached/Other dwelling at 4/22-28 O'Brien Street, which is over-predicted by about 65.9%. These cases suggest the model is averaging toward more common properties when a dwelling has unusual prestige or attached-property characteristics. Information such as ocean outlook, architectural quality, renovation, frontage, development potential, exact street position and sale conditions could materially improve these predictions.

Predictions should therefore be interpreted cautiously for prestige homes, unusual attached dwellings, development sites and any property far from the training distribution. A numerical prediction can look precise even when the underlying feature representation is incomplete.

5. Human judgement, machine learning and an LLM

Part 5 reproducibility protocol

Selection: sort the untouched test set alphabetically by address within suburb, then take 4 Bondi Beach, 3 Parramatta and 3 Blacktown properties. This produces the same 10 rows every time and does not inspect target price or model error. The exact LLM prompt is stored in part5_llm_prompt_output_log.md; blinded feature-only inputs are stored in part5_blinded_inputs.csv.

Ten properties were selected from the untouched held-out test set using a deterministic rule fixed independently of price and prediction error. Within each suburb, test rows were sorted alphabetically by address; the first four Bondi Beach rows, first three Parramatta rows, and first three Blacktown rows were selected. The ML prediction came from the selected Random Forest. The LLM valuation used only suburb, property type, bedrooms, bathrooms, parking, land size and sale date. A transparent human-style comparable baseline started from the training-set median for the same suburb/property type and applied simple bedroom, bathroom, parking and land adjustments.

Property	Actual	ML	LLM	Manual baseline
10 Forest Knoll Avenue	A$8,000,000	A$5,221,356	A$6,200,000	A$5,450,000
160 Hall Street	A$4,450,000	A$3,688,567	A$4,100,000	A$4,150,000
29 Sir Thomas Mitchell Road	A$5,800,000	A$3,330,239	A$4,400,000	A$3,700,000
4/22-28 O'Brien Street	A$1,670,000	A$2,770,971	A$1,800,000	A$1,500,000
1/77-79 Thomas Street	A$1,300,000	A$1,232,309	A$1,350,000	A$1,250,000
10/43 Pemberton Street	A$1,101,500	A$1,168,630	A$1,150,000	A$1,150,000
2/25-29 O'Connell Street	A$851,000	A$1,127,324	A$950,000	A$1,000,000
1 Tara Road	A$820,000	A$951,870	A$900,000	A$950,000
12 Siebel Street	A$1,080,000	A$1,288,750	A$1,350,000	A$1,250,000
24 Twilight Crescent	A$1,024,000	A$1,107,587	A$1,100,000	A$1,150,000

Approach	MAE	RMSE
ML model	A$794,616	A$1,255,623
LLM-style estimate	A$430,350	A$737,505
Manual human-style baseline	A$579,350	A$1,054,520

In the recorded ten-property comparison, the feature-only LLM estimate has the lowest MAE (A$430,350), followed by the manual baseline (A$579,350) and the ML model (A$794,616). This small comparison is highly sensitive to a few difficult Bondi Beach properties and should not be generalised beyond the selected sample. The ML model is consistent, reproducible and tied to observed training data; the LLM can apply broad contextual priors but is not guaranteed to be current or calibrated; human judgement can incorporate qualitative attributes but may be subjective.

Reproducibility evidence is included in the submission archive. `part5_llm_prompt_output_log.md` records the exact prompt template, the 10 property inputs and the numeric LLM estimates used by the notebook; `part5_blinded_inputs.csv` contains the same 10 inputs with no sale-price column. The selection rule uses only suburb and address. The original integrated GenAI workflow did not preserve a complete raw chat transcript or a provably blinded session record, so I do not retrospectively claim that

the historical LLM estimates can be proven to have been generated before the assistant had access to actual prices. For the final marked comparison, the included blinded-rerun checklist should be followed in a fresh LLM session before revealing any actual prices, and the raw outputs should be retained verbatim.

6. Deployment and reflection

The final fitted preprocessing-and-model pipeline was saved with joblib and connected to a Gradio application. The single-property interface accepts suburb, property type, bedrooms, bathrooms, parking, land size and valuation date, then returns an estimated sale price. A second tab accepts batch CSV uploads using the same fields. Keeping preprocessing inside the fitted pipeline reduces deployment mismatch because imputation and one-hot encoding are applied exactly as they were during model training.

The submission archive contains the dataset, executed notebook, `run_analysis.py`, `create_dataset.py`, `best_model.joblib`, `app.py`, `requirements.txt`, sample batch input, figures, and the Part 5 reproducibility files. An externally accessible GitHub/OneDrive/Dropbox URL still needs to be inserted before final OnTrack submission.

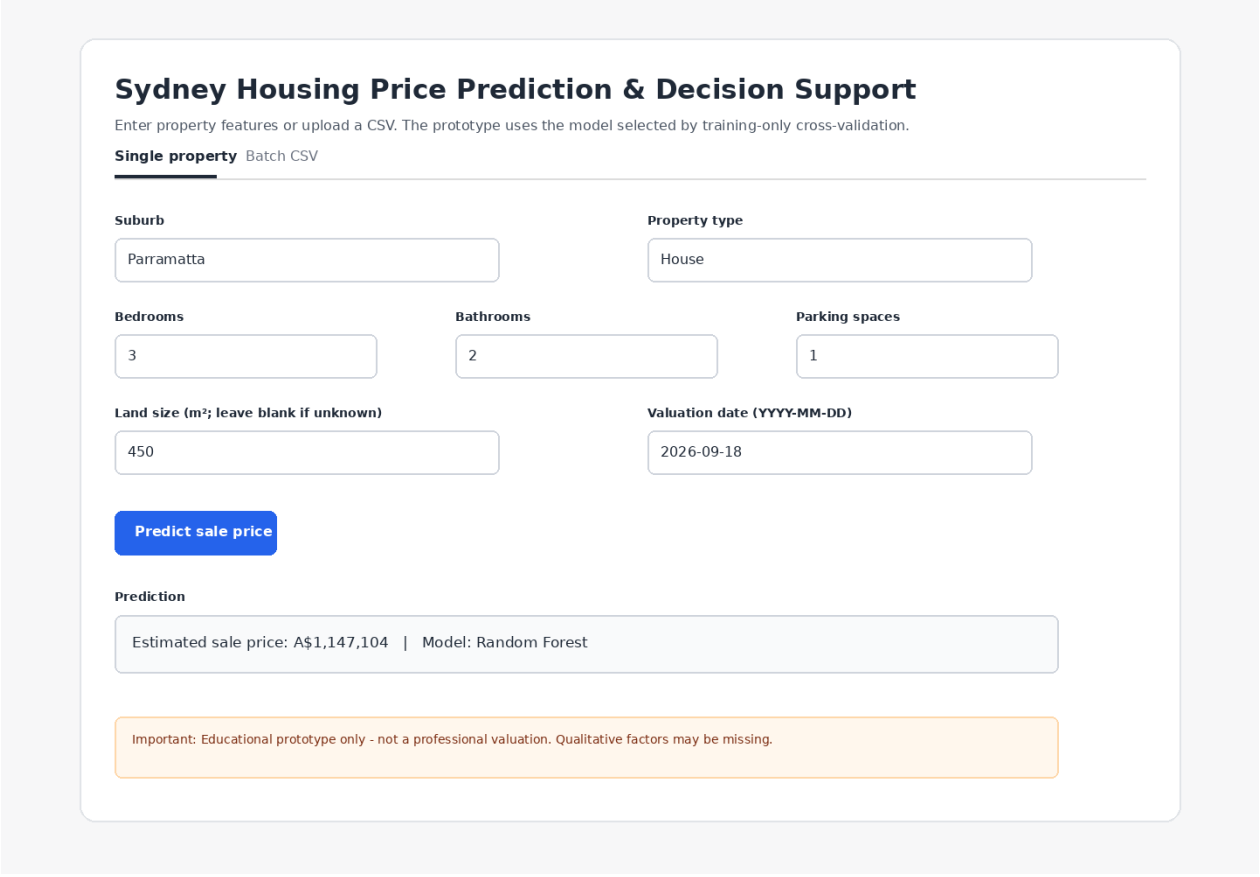


Figure 8. Gradio application interface preview using an actual model prediction.

Application instructions

1. Place app.py, best_model.joblib and requirements.txt in the same folder.
2. Create and activate a Python virtual environment.
3. Run: pip install -r requirements.txt

4. Run: python app.py
5. Open http://127.0.0.1:7860 in a browser.
6. For a single prediction, enter property features and select Predict sale price. For batch prediction, upload a CSV with suburb, property_type, bedrooms, bathrooms, parking, land_size_m2 and sale_date.

Critical reflection

The main lesson is that data coverage matters more than model complexity. Even good cross-validation scores cannot recover information that was never collected. A larger and more recent dataset with floor area, exact coordinates, distance to transport/coast, property age, school catchment, zoning, renovation condition, views and agent descriptions would likely reduce the largest errors. Text embeddings or image features could capture some qualitative factors, while repeated or nested cross-validation could provide more stable model-selection estimates.

Automated valuation also has ethical and fairness implications. Historical sale data can encode neighbourhood-level inequality and may underrepresent unusual property types. Users may treat a precise dollar estimate as objective even when the model is uncertain. The prototype therefore describes its output as decision support, not a professional valuation, and highlights missing qualitative factors. A production system should add uncertainty intervals, drift monitoring, documented data provenance and human review for high-impact decisions.

GenAI acknowledgement

ChatGPT (OpenAI, GPT-5.6 Sol) was used to assist with planning the workflow, drafting and debugging Python code, organising the report, and supporting the feature-only LLM comparison. The property records were transcribed from the cited public sold-listing pages and the analysis outputs were produced by the supplied executable notebook/scripts. The exact Part 5 prompt template and recorded numeric estimates are included in the archive; the report explicitly distinguishes the original integrated workflow from a strictly blinded rerun. The student remains responsible for critically reviewing, understanding and editing the final submission.

References

[1] SIT720 8.1D assessment brief, "Sydney Housing Price Prediction and Decision Support System," Deakin University, 2026.

[2] Realestate.com.au, Sold houses/townhouses in Bondi Beach NSW 2026, pages 1-2, accessed 18 Sep. 2026. https://www.realestate.com.au/sold/property-house-townhouse-in-bondi%2Bbeach,%2Bnsw%2B2026/list-1

[3] Realestate.com.au, Sold houses/land/townhouses/acreage in Parramatta NSW 2150, pages 1-2, accessed 18 Sep. 2026. https://www.realestate.com.au/sold/property-house-land-townhouse-acreage-in-parramatta,%2Bnsw%2B2150/list-1

[4] Realestate.com.au, Sold houses/villas in Blacktown NSW 2148, pages 1-2, accessed 18 Sep. 2026. https://www.realestate.com.au/sold/property-house-villa-in-blacktown,%2Bnsw%2B2148/list-1

[5] Scikit-learn developers, "scikit-learn: Machine Learning in Python," documentation. https://scikit-learn.org/stable/

[6] Gradio, "Gradio Documentation." https://www.gradio.app/docs